Arabic Policy Analyzer

RAG

Multi-Document Agentic RAG System for Arabic and English Policy PDFs

Problem and Motivation

Policy and regulation PDFs are difficult to search because the answer often depends on both exact wording and surrounding legal context.

Long documents

Large PDFs with formal sections, clauses, and repeated boilerplate.

Arabic complexity

RTL text, inconsistent character forms, and high-context legal language.

Exact terms matter

Definitions, article names, controls, and regulatory expressions must be preserved.

Need traceability

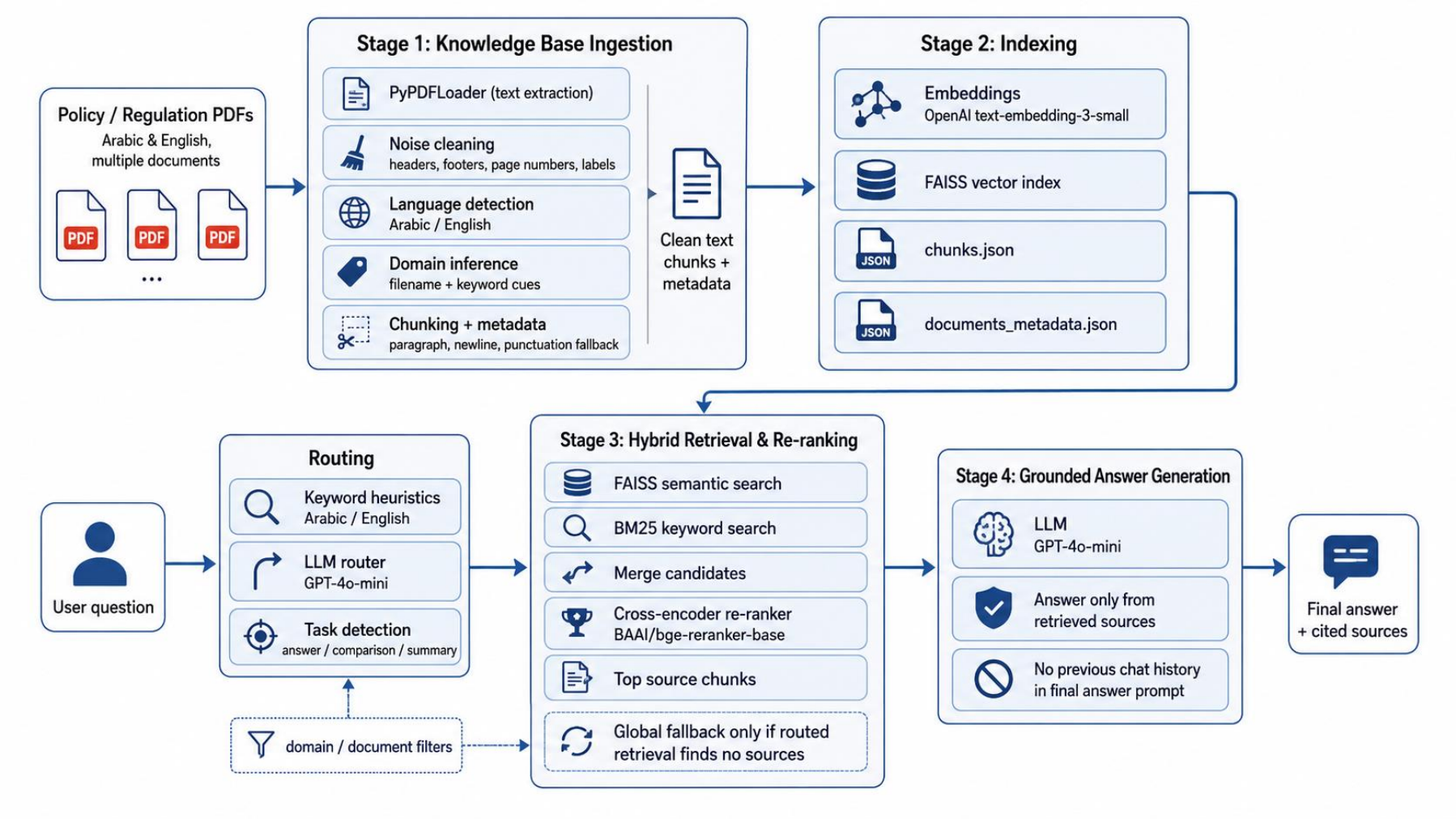
Answers should be grounded in retrieved sources, not general model knowledge.

Project Objective

The goal is to help users ask questions across multiple Arabic and English policy documents and receive grounded answers with source references.

- Support several regulatory domains rather than one static PDF.
- Combine semantic search with exact keyword search.
- Use a cross-encoder reranker before generation.
- Generate answers only from selected source chunks.

System Architecture



CORE IMPLEMENTATION STACK

AI Tools and Technologies Used

Streamlit	Chat interface, sidebar controls, and rebuild action
LangChain + PyPDFLoader	Document loading and chunk preparation workflow
OpenAI text-embedding-3-small	Dense embeddings for semantic retrieval
FAISS	Vector index for fast similarity search
BM25	Keyword retrieval for exact policy terms
BAAI/bge-reranker-base	Cross-encoder re-ranking for better precision
GPT-4o-mini	Routing, task detection, and grounded answer generation

CURRENT KNOWLEDGE BASE COVERAGE

Indexed Document Collection

The current knowledge base contains four policy and regulation PDFs from SDAIA and NCA, covering 105 pages and 152 retrievable chunks.

4 Documents

105 Pages

152 Chunks

3 Arabic + 1 English Bilingual policy analysis

File	Domain	Lang	Authority	Pages	Chunks
CriticalSystemsCybersecurityControls.pdf	Critical Systems Cybersecurity	AR	NCA	32	37
DataSharingPolicyAR.pdf	Data Sharing	AR	SDAIA	17	19
EssentialCybersecurityControls-NCA.pdf	Cybersecurity Controls	AR	NCA	40	67
PersonalDataProtectionLaw-English.pdf	Personal Data Protection	EN	SDAIA	16	29

Chunking and Indexing Configuration

Chunking balances context and precision: chunks must be large enough to preserve policy meaning but small enough to retrieve relevant evidence accurately.

RecursiveCharacterTextSplitter

chunk_size = 1800 characters
chunk_overlap = 250 characters
keep_separator = True
separators = GENERIC_SEPARATORS

Why 1,800 characters?
A moderate size keeps enough surrounding context for definitions, rules, exceptions, and controls without making retrieval too broad.

Why 250 overlap?
Overlap protects continuity when an article, clause, or requirement crosses a chunk boundary.

Why keep separators?
Separators help preserve structural markers such as paragraphs, punctuation, and section boundaries, which are important in policy documents.

Final index: 152 chunks stored as FAISS vectors + readable chunks in chunks.json

Evaluation Results

The proposed pipeline was evaluated against a simpler baseline to measure whether routing, hybrid retrieval, and re-ranking improved answer quality.

Groundedness
3.0 → 8.7

Correctness
3.3 → 5.3

Main takeaway
The proposed system produced answers that were much more strongly supported by retrieved source content.

Metric	Simple-RAG Baseline	Proposed Agentic RAG
Groundedness	3.0	**8.7**
Correctness	3.3	**5.3**

- Routing narrowed the search space to the most relevant domain or document.
- Hybrid retrieval improved coverage by combining semantic and exact-term matching.
- Cross-encoder re-ranking improved precision before the LLM generated the answer.

Conclusion

Any Question?

RAG